

Derivation of All Mamba-3 Equations

This article derives the key equations in the Mamba-3 paper with one goal: **every single step should be easy to follow**. No steps are skipped. Each equation transforms into the next with a clear reason. If you can follow basic calculus and linear algebra, you can follow this entire derivation.

1 Definitions: Dimensions and Parameters

1.1 Basic Dimensions

- T : Sequence Length
- D : Number of Input/Output Channels (Model Dimension)
- N : State Dimension (per channel)

1.2 Variable Dimension Definitions

- $\mathbf{X} \in \mathbb{R}^{T \times D}$: Input sequence
- $\mathbf{Y} \in \mathbb{R}^{T \times D}$: Output sequence
- $\mathbf{h}(t) \in \mathbb{C}^{D \times N}$: State matrix. Each column $\mathbf{h}_d(t) \in \mathbb{C}^{N \times 1}$ is the state vector of channel d .
- $\mathbf{A}(t)$: Collection of diagonal components.
 - Storage dimension: $\mathbb{C}^{D \times N}$. Each element $A_{d,n}(t)$ is the complex coefficient for the n -th state of channel d .
 - Operational dimension: $\mathbb{C}^{D \times N \times N}$. For each channel d , it acts as an $N \times N$ diagonal matrix $\mathbf{A}_d(t) = \text{diag}(A_{d,1}(t), \dots, A_{d,N}(t))$.
- $\mathbf{B}(t) \in \mathbb{R}^{T \times N}$, $\mathbf{C}(t) \in \mathbb{R}^{T \times N}$: Projection vectors shared across all channels.

2 Derivation of Equation (1): The Discrete SSM Recurrence

The paper's equation (1) is the discrete-time recurrence:

$$\mathbf{h}_t = \alpha_t \mathbf{h}_{t-1} + \gamma_t \mathbf{B}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t^\top \mathbf{h}_t \quad (\text{paper 1})$$

where:

- $A_t < 0$ is a scalar (the *scalar SSM* parameterization: $\mathbf{A}_t = A_t \mathbf{I}_N$). The constraint $A_t < 0$ ensures stability: in the continuous-time ODE, the homogeneous solution is $\mathbf{h}(t) \propto e^{A_t t}$, which decays only when $A < 0$. If $A > 0$, the state would grow exponentially.
- $\Delta_t > 0$ is the time step size.
- $\alpha_t := e^{\Delta_t A_t} \in (0, 1)$ is the scalar state-transition.

- $\gamma_t := \Delta_t$ is the discretization factor.
- $\mathbf{B}_t \in \mathbb{R}^N$: input projection, $\mathbf{C}_t \in \mathbb{R}^N$: output projection, $\mathbf{h}_t \in \mathbb{R}^N$: hidden state.

We derive this from the continuous-time state space model (SSM).

2.1 Continuous-Time SSM

The underlying continuous-time ordinary differential equation (ODE) is:

$$\dot{\mathbf{h}}(t) = \mathbf{A}(t) \mathbf{h}(t) + \mathbf{B}(t) x(t), \quad y(t) = \mathbf{C}(t)^\top \mathbf{h}(t) \quad (1)$$

where $\mathbf{h}(t) \in \mathbb{R}^N$, $\mathbf{A}(t) \in \mathbb{R}^{N \times N}$, $\mathbf{B}(t), \mathbf{C}(t) \in \mathbb{R}^N$, and $x(t), y(t) \in \mathbb{R}$. This is for a single channel; the multi-channel (MIMO) extension is discussed in Section 3.

2.2 Continuous-Time Solution (Integrating Factor Method)

To solve the first-order linear non-homogeneous ODE (1), we multiply both sides by the integrating factor $\exp(-\int_0^t \mathbf{A}(\tau) d\tau)$:

$$\exp\left(-\int_0^t \mathbf{A}(\tau) d\tau\right) \left[\dot{\mathbf{h}}(t) - \mathbf{A}(t) \mathbf{h}(t) \right] = \exp\left(-\int_0^t \mathbf{A}(\tau) d\tau\right) \mathbf{B}(t) x(t) \quad (2)$$

We apply the reverse of the product rule $\frac{d}{dt}(fg) = f'g + fg'$ to the left-hand side:

$$\frac{d}{dt} \left[\exp\left(-\int_0^t \mathbf{A}(\tau) d\tau\right) \mathbf{h}(t) \right] = \exp\left(-\int_0^t \mathbf{A}(\tau) d\tau\right) \mathbf{B}(t) x(t) \quad (3)$$

Setting the initial state $\mathbf{h}(0) = \mathbf{0}$ and integrating both sides from 0 to t :

$$\exp\left(-\int_0^t \mathbf{A}(\tau) d\tau\right) \mathbf{h}(t) - \underbrace{\exp(0) \mathbf{h}(0)}_{=\mathbf{0}} = \int_0^t \exp\left(-\int_0^s \mathbf{A}(\tau) d\tau\right) \mathbf{B}(s) x(s) ds \quad (4)$$

Multiplying both sides from the left by $\exp(\int_0^t \mathbf{A}(\tau) d\tau)$:

$$\mathbf{h}(t) = \exp\left(\int_0^t \mathbf{A}(\tau) d\tau\right) \int_0^t \exp\left(-\int_0^s \mathbf{A}(\tau) d\tau\right) \mathbf{B}(s) x(s) ds \quad (5)$$

Since $\exp(\int_0^t \mathbf{A} d\tau)$ does not depend on s , we bring it inside the integral by linearity. Then, because $\mathbf{A}$ is diagonal, $\int_0^t \mathbf{A} d\tau$ and $\int_0^s \mathbf{A} d\tau$ commute, so we can combine the two matrix exponentials via $\exp(\mathbf{M}_1) \exp(\mathbf{M}_2) = \exp(\mathbf{M}_1 + \mathbf{M}_2)$:

$$\mathbf{h}(t) = \int_0^t \exp\left(\int_0^t \mathbf{A}(\tau) d\tau - \int_0^s \mathbf{A}(\tau) d\tau\right) \mathbf{B}(s) x(s) ds \quad (6)$$

By the linearity of integration $\int_0^t - \int_0^s = \int_s^t$, we obtain the continuous-time general solution:

$$\mathbf{h}(t) = \int_0^t \underbrace{\exp\left(\int_s^t \mathbf{A}(\tau) d\tau\right)}_{\Phi(t,s)} \mathbf{B}(s) x(s) ds \quad [\mathbf{h}(t) \in \mathbb{R}^N] \quad (7)$$

2.3 Discretization

We derive the state $\mathbf{h}_t$ at time τ_t by splitting the integration range $[0, \tau_t]$ into $[0, \tau_{t-1}]$ and $[\tau_{t-1}, \tau_t]$:

$$\mathbf{h}_t = \int_0^{\tau_{t-1}} \Phi(\tau_t, s) \mathbf{B}(s) x(s) ds + \int_{\tau_{t-1}}^{\tau_t} \Phi(\tau_t, s) \mathbf{B}(s) x(s) ds \quad (8)$$

We apply the semigroup property $\Phi(t, s) = \Phi(t, u) \Phi(u, s)$ to the first term:

$$\mathbf{h}_t = \Phi(\tau_t, \tau_{t-1}) \underbrace{\int_0^{\tau_{t-1}} \Phi(\tau_{t-1}, s) \mathbf{B}(s) x(s) ds}_{\mathbf{h}_{t-1}} + \int_{\tau_{t-1}}^{\tau_t} \Phi(\tau_t, s) \mathbf{B}(s) x(s) ds \quad (9)$$

Right-hand approximation. Following the paper (Section 3.1.1), we approximate $\mathbf{A}(s) \approx \mathbf{A}(\tau_t) =: \mathbf{A}_t$ for all $s \in [\tau_{t-1}, \tau_t]$, with $\Delta_t := \tau_t - \tau_{t-1}$. This gives: **State-transition.** From the definition of Φ in (7):

$$\Phi(\tau_t, \tau_{t-1}) = \exp\left(\int_{\tau_{t-1}}^{\tau_t} \mathbf{A}(s) ds\right) \approx \exp\left(\int_{\tau_{t-1}}^{\tau_t} \mathbf{A}_t ds\right) = \exp(\mathbf{A}_t (\tau_t - \tau_{t-1})) = \exp(\Delta_t \mathbf{A}_t) \quad (10)$$

State-input integral. Starting from the second term in (9), we expand $\Phi(\tau_t, s)$ under the right-hand approximation $\mathbf{A}(s) \approx \mathbf{A}_t$:

$$\begin{aligned} \int_{\tau_{t-1}}^{\tau_t} \Phi(\tau_t, s) \mathbf{B}(s) x(s) ds &= \int_{\tau_{t-1}}^{\tau_t} \exp\left(\int_s^{\tau_t} \mathbf{A}(\sigma) d\sigma\right) \mathbf{B}(s) x(s) ds \\ &\approx \int_{\tau_{t-1}}^{\tau_t} \exp((\tau_t - s) \mathbf{A}_t) \mathbf{B}(s) x(s) ds \end{aligned} \quad (11)$$

Under exponential-Euler, we further hold $\mathbf{B}(s) \approx \mathbf{B}_t$ and $x(s) \approx x_t$:

$$\approx \int_{\tau_{t-1}}^{\tau_t} \exp((\tau_t - s) \mathbf{A}_t) ds \mathbf{B}_t x_t \approx \Delta_t \mathbf{B}_t x_t \quad (12)$$

This yields the general discrete update:

$$\mathbf{h}_t \approx \exp(\Delta_t \mathbf{A}_t) \mathbf{h}_{t-1} + \Delta_t \mathbf{B}_t x_t \quad (13)$$

Scalar SSM parameterization. Mamba-2/3 parameterizes $\mathbf{A}_t = A_t \mathbf{I}_N$ with $A_t < 0$ (scalar times identity). Defining

$$\alpha_t := e^{\Delta_t A_t} \in (0, 1), \quad \gamma_t := \Delta_t \quad (14)$$

the update (13) becomes:

$$\boxed{\mathbf{h}_t = \alpha_t \mathbf{h}_{t-1} + \gamma_t \mathbf{B}_t x_t, \quad y_t = \mathbf{C}_t^\top \mathbf{h}_t} \quad (15)$$

which is the paper's equation (1). ■

2.4 Recursive Unrolling

We unroll the recurrence (15) starting from $\mathbf{h}_0 = \mathbf{0}$.

Base cases.

$$\begin{aligned}
\mathbf{h}_0 &= \mathbf{0} \\
\mathbf{h}_1 &= \alpha_1 \mathbf{h}_0 + \gamma_1 \mathbf{B}_1 x_1 = \gamma_1 \mathbf{B}_1 x_1 \\
\mathbf{h}_2 &= \alpha_2 \mathbf{h}_1 + \gamma_2 \mathbf{B}_2 x_2 = \alpha_2 \gamma_1 \mathbf{B}_1 x_1 + \gamma_2 \mathbf{B}_2 x_2 \\
\mathbf{h}_3 &= \alpha_3 \mathbf{h}_2 + \gamma_3 \mathbf{B}_3 x_3 = \alpha_3 \alpha_2 \gamma_1 \mathbf{B}_1 x_1 + \alpha_3 \gamma_2 \mathbf{B}_2 x_2 + \gamma_3 \mathbf{B}_3 x_3
\end{aligned} \tag{16}$$

Pattern. Each term indexed by s carries the input $\gamma_s \mathbf{B}_s x_s$ propagated forward through the product of transition scalars from step $s+1$ to t . Using the paper’s cumulative product notation $\alpha_{t:s}^\times := \prod_{j=s}^t \alpha_j$ (with $\alpha_{t:t+1}^\times = 1$ when the range is empty):

$$\mathbf{h}_t = \sum_{s=1}^t \alpha_{t:s+1}^\times \gamma_s \mathbf{B}_s x_s \tag{17}$$

3 Derivation of Equation (2): The Matrix Output Form

Using the discrete SSM derived in Section 2, we now derive the matrix output form $\mathbf{Y} = (\mathbf{L} \odot \mathbf{CB}^\top) \mathbf{X}$ (paper equation 2).

3.1 Output Equation

The output at time step t is:

$$y_t = \mathbf{C}_t^\top \mathbf{h}_t \tag{18}$$

Substituting the unrolled state (17):

$$y_t = \mathbf{C}_t^\top \sum_{s=1}^t \alpha_{t:s+1}^\times \gamma_s \mathbf{B}_s x_s = \sum_{s=1}^t \alpha_{t:s+1}^\times \gamma_s (\mathbf{C}_t^\top \mathbf{B}_s) x_s \tag{19}$$

Since α_t is scalar, the cumulative product $\alpha_{t:s+1}^\times$ commutes freely with the vectors.

3.2 Kernel Factorization

Define (for $t, s = 1, \dots, T$):

- Structured mask $\mathbf{L} \in \mathbb{R}^{T \times T}$: (t, s) -element $L_{ts} = \alpha_{t:s+1}^\times \gamma_s = (\prod_{j=s+1}^t \alpha_j) \gamma_s$ (lower triangular: $L_{ts} = 0$ for $s > t$, because the recurrence is causal — the unrolled sum (17) only includes inputs up to time t , so future inputs x_s with $s > t$ have no contribution)
- Projection kernel $\mathbf{CB}^\top \in \mathbb{R}^{T \times T}$: (t, s) -element $(\mathbf{CB}^\top)_{ts} = \mathbf{C}_t^\top \mathbf{B}_s = \sum_{n=1}^N C_{t,n} B_{s,n}$

Note that $\mathbf{L}$ absorbs both the cumulative transition $\alpha_{t:s+1}^\times$ and the discretization factor γ_s . Both $\mathbf{L}$ and $\mathbf{CB}^\top$ are $T \times T$ matrices (shared across channels).

The output becomes:

$$y_t = \sum_{s=1}^T L_{ts} \cdot (\mathbf{CB}^\top)_{ts} \cdot x_s = \sum_{s=1}^T (\mathbf{L} \odot \mathbf{CB}^\top)_{ts} x_s \tag{20}$$

3.3 Multi-Channel (MIMO) Extension

The recurrence (15) is applied independently to each of D channels with the same $\mathbf{B}_t$, $\mathbf{C}_t$, and $\mathbf{L}$, but potentially different inputs $x_t^{(d)}$. Collecting all channels into matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{T \times D}$, the per-channel outputs stack into:

$$\boxed{\mathbf{Y} = (\mathbf{L} \odot \mathbf{C}\mathbf{B}^\top) \mathbf{X}} \quad (21)$$

where $(\mathbf{L} \odot \mathbf{C}\mathbf{B}^\top) \in \mathbb{R}^{T \times T}$ is a single matrix applied to all D columns of $\mathbf{X}$. ■